

Nourhan Bayasi, Ph.D.

AI Scientist

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About Me

I develop AI systems for healthcare, focusing on solutions that can be translated into real clinical practice. My background spans both academia and industry, allowing me to approach problems with both rigor and practicality. I am passionate about collaborative work and actively mentor junior researchers, helping them build strong and critical thinking skills.

Professional Experience

Postdoctoral Research Fellow **Vancouver, Canada**
BC Cancer | UBC Faculty of Medicine *07/2025 - present*

- Developing physician-in-the-loop and agentic AI systems for CT and PET workflows.
- Designing interactive AI systems integrating clinician feedback to improve robustness and deployment readiness.
- Translating research into clinically applicable AI solutions.

Founder and Mentor **Online**
Rising Scholars Hub (RiSH) *03/2025 - present*

- Founded a global mentorship initiative supporting early-career researchers in AI and healthcare.
- Organized workshops and collaborative research activities.
- Built an international research network across disciplines.

Graduate Teaching Assistant **Vancouver, Canada**
University of British Columbia (UBC) *09/2020 - 04/2025*

- Delivered instruction across labs, tutorials, and office hours.
- Evaluated assignments, exams, and projects with consistent academic standards.
- Coordinated lab logistics and setup.

Machine Learning Engineer Intern **Vancouver, Canada**
Cognia AI *04/2024 - 10/2024*

- Developed bias mitigation algorithms to improve fairness in ML systems.
- Enhanced continual learning pipelines for evolving data.
- Contributed to trustworthy AI development practices.

Graduate Academic Assistant **Vancouver, Canada**
University of British Columbia (UBC) *06/2020 - 08/2020*

- Designed online course materials and assessments.
- Integrated digital tools to improve engagement.

Lab Instructor, Electrical Engineering **Sharjah, UAE**
Higher Colleges of Technology (HCT) *08/2017 - 12/2019*

- Delivered lab instruction and supervised student projects.
- Managed lab operations and safety systems.
- Mentored students to six top-three national competition wins.

Workshop Engineer, Electrical Engineering **Umm Al Quwain, UAE**
Institute of Applied Technology (IAT) *09/2015 - 07/2017*

- Supported curriculum delivery and student projects.
- Mentored students to multiple first-place science fair wins.
- Managed lab setup, maintenance, and procurement.

Academic Education

PhD in Electrical and Computer Engineering **Vancouver, Canada**
University of British Columbia (UBC) *01/2020 – 03/2025*

- Thesis: *Beyond Catastrophic Forgetting: Advancing Continual Learning for Robust and Fair Medical Image Analysis*
- CGPA: 94.1%

Master of Science in Electrical and Computer Engineering **Abu Dhabi, United Arab Emirates**
Khalifa University (KU) *09/2013 – 04/2015*

- Thesis: *An ECG signal processor for the Prediction of ventricular arrhythmia*
- CGPA: 3.7/4

Bachelor of Science in Communication Engineering
Khalifa University (KU)
 – CGPA: 3.96/4 (1st in cohort)

Sharjah, United Arab Emirates
 09/2009 – 04/2013

Technical Skills

Software Programming

Python, MATLAB, C++

Hardware Programming

Synopsys Custom Flow, Verilog, SystemVerilog

Simulation

Tinkercad, Multisim, Simulink

Honors and Awards

Research Awards and Recognition

- **3rd Place, Best Paper Award, EMERGE Workshop @ MICCAI** 2025
- **1st Place, WiM Best Health Equity Paper Award, MICCAI Conference** 2024
- **2nd Place, WiM Best Oral Presentation Award, MICCAI Conference** 2024
- **Shortlisted, MICCAI Best Paper Award, MICCAI Conference** 2024
- **Shortlisted, MICCAI Young Scientist Award, MICCAI Conference** 2024
- **Best Paper Award, ISIC Medical Image Analysis Workshop @ MICCAI** 2023
- **Best Paper Award, ISIC Medical Image Analysis Workshop @ ECCV** 2022
- **Best Paper Award, IEEE Transactions on Very Large-Scale Integration Systems** 2016

Fellowships and Scholarships

- **Canada Postdoctoral Research Award (CPRA), NSERC** 2026-2028
- **Vanier Canada Graduate Scholarship (Ranked Top 1 Nationwide), NSERC** 2022-2025
- **Borealis AI Global Fellowship, RBC's AI Research Institute** 2024
- **Four Year Doctoral Fellowship (4YF), University of British Columbia (UBC)** 2022-2026
- **International Student Award, University of British Columbia (UBC)** 2020-2022
- **Scholarship Award, PhD Studies, University of British Columbia (UBC)** 2020-2023
- **Scholarship Award, Master Studies, Khalifa University (KU)** 2013-2015
- **Scholarship Award, Bachelor Studies, Khalifa University (KU)** 2009-2013

Academic and Institutional Awards

- **Dr. and Mrs. Brandwajn Graduate Award, University of British Columbia (UBC)** 2021
- **Faculty of Applied Science Graduate Award, University of British Columbia (UBC)** 2021
- **President's Academic Excellence Initiative PhD Award, University of British Columbia (UBC)** 2021
- **Society Registration Grant, MICCAI Conference** 2024
- **Student Travel Grant, MICCAI Conference** 2021
- **Best Prototype Award, 2nd Place, Engineering Student Renewable Energy Competition @ UAE University** 2013
- **Best Poster Award, 1st Place, Undergraduate Research Conference on Applied Computing @ Zayed University** 2013
- **Leadership Award, Best Student Category, Khalifa University (KU)** 2013
- **President List Award, College of Engineering, Khalifa University (KU)** 2009-2013

Leadership and Service

- **Founder and Organizer, CLiMeM Workshop @ MICCAI** 2026 – Current
- **Judge, AI and Machine Learning Session, UBC Radiology Clinical & Fundamental Science Research Day** 2026
- **Public Outreach and Membership Officer, Women in MICCAI (WiM)** 2025 – Current
- **Doctoral Programs Officer, MICCAI Student Board** 2025 – Current
- **Electronics Expert, Emirates Skills National Competition, UAE** 2017
- **Standard Leader, AdvancED Academic Accreditation, Institute of Applied Technology, UAE** 2016
- **STEAM Program Developer, Engineering, Institute of Applied Technology, UAE** 2016
- **Chairman, IEEE Khalifa University Student Branch, UAE** 2010

Professional Activities

- **Reviewer**, *MICCAI, ISIC Skin Image Analysis Workshop @ MICCAI* 2022 – Current
- **Reviewer**, *ECCV, NeurIPS, IEEE TMI* 2026
- **Reviewer & Emergency Reviewer**, *MICCAI, WACV, MSB EMERGE @ MICCAI, MedIA, IEEE TNNLS (A/E)* 2025
- **Reviewer**, *CVPR, Medical Image Analysis (MedIA), IEEE TMI* 2024

Personal Information

- **Nationality**: *Canadian*
- **Date of Birth**: *15 January 1992*